* Batch Vs Mini-Batch vs SGD.

All three are ways to compute Gradients and update model weight during training.

Dataset → Choose a Training Method → Compute Gradient → Update weight → Repeat

@codingdidi

① Batch Gradient Descent:

⇒ Use the Entire Dataset to compute Gradient once per update.

□ □ □ □ □ □ □ □ → One Update

All data (N sample)

How it works

- Take all N sample
- Compute loss
- Compute Gradient
- Update weight once
- Repeat

$\boxed{\text{Dataset} = 1000 \text{ Samples} \longrightarrow 1 \text{ update per epoch.}}$

② Mini - Batch

⇒ Use a small Batch of data to compute gradient and update. More common in practice.

Batch 1	Batch 2	...	Batch K
↓	↓		↓
Update 1	Update 2		Update 3

How it works?

* Split data into small batch of size B.
* for each batch:
  - Compute loss
  - Compute Gradient
  - update Weights
* Repeat for all batches

Dataset = 1000 Samples, Batch size (B) = 100 $\longrightarrow \frac{1000}{100} = 10$ updates per epoch.

③ Stochastic Gradient Descent (SGD)

⇒ Use One sample at a time to compute

Sample 1	Sample 2	Sample 3	.....	Sample N
□	□	□	.....	□
↓	↓	↓		↓
Update 1	Update 2	Update 3		Update N

How it works?

* pick one sample
* Compute loss
* Compute Gradient
* Update Weight.
* Repeat

In Deep Learning, Mini - Batch Gradient Descent is widely used!